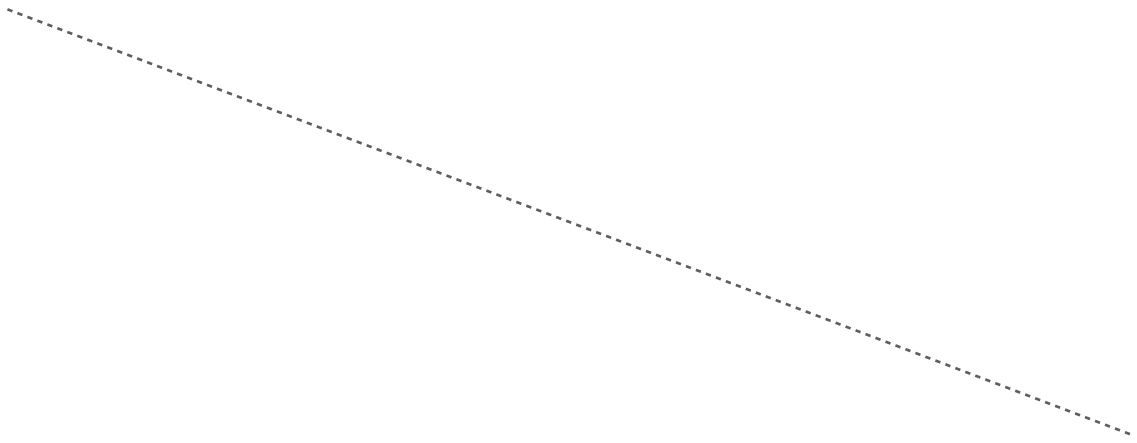
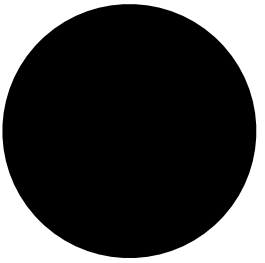
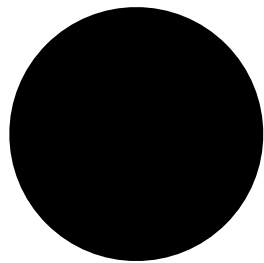
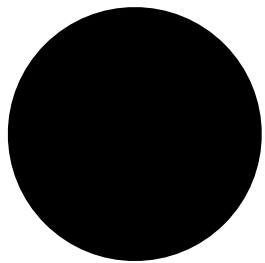
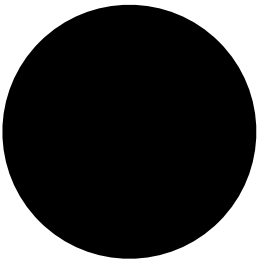
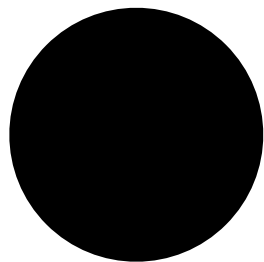
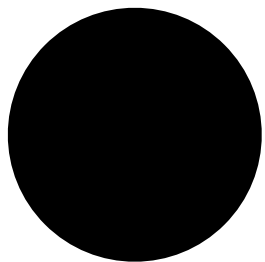
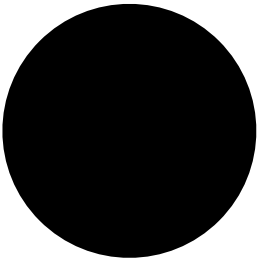
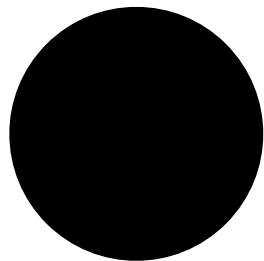
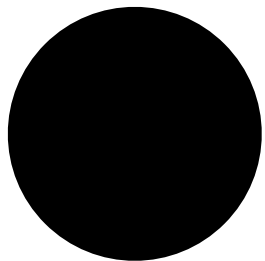
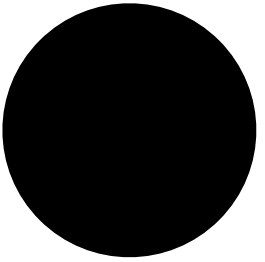
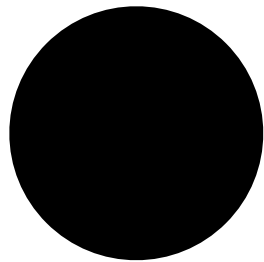
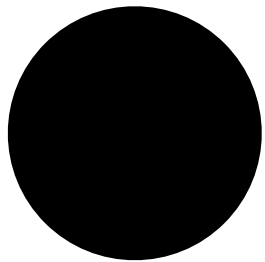










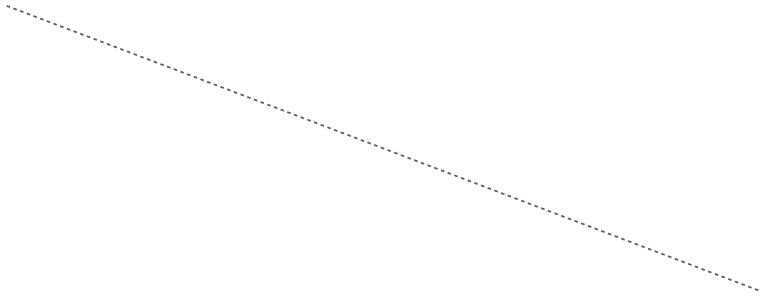
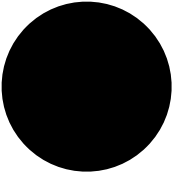
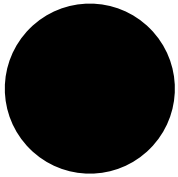
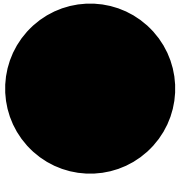
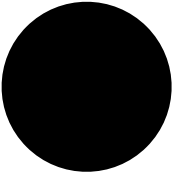
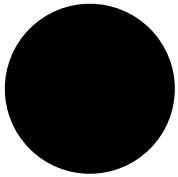
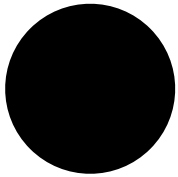
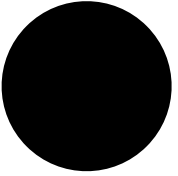
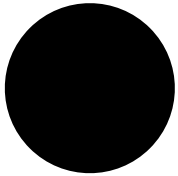
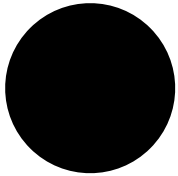
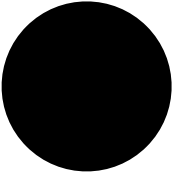
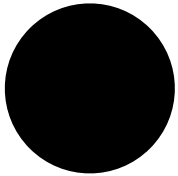
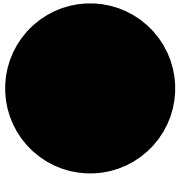












Constructing  $J$

---

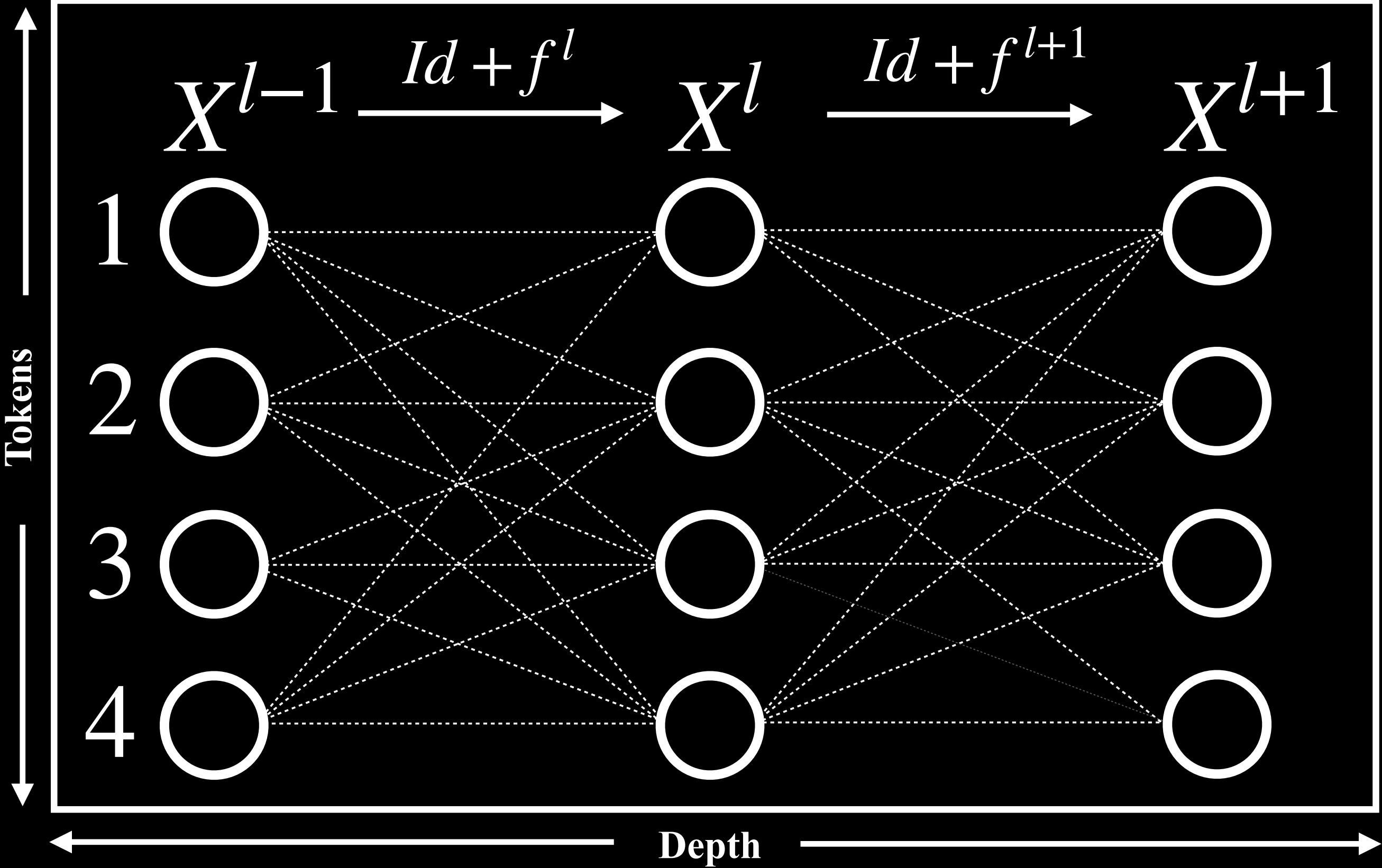

$$J(X) = \frac{\partial f}{\partial X}(X)$$

Constructing  $A$

---


$$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

$$A = \tilde{U}^T J \tilde{V}$$



$\tilde{U} \overset{?}{\sim} U$ 
 $\tilde{V} \overset{?}{\sim} V$

$\downarrow$ 
 $\downarrow$

$\tilde{U}^T U \overset{?}{=} I$ 
 $\tilde{V}^T V \overset{?}{=} I$

$$A = \tilde{U}^T J \tilde{V}$$

$$= \tilde{U}^T U S V^T \tilde{V}$$

$$\overset{?}{=} S$$

$$m(A) = \frac{\|A - S\|_F}{\|S\|_p}$$

Constructing  $J$

Constructing  $A$

$$J(X) = \frac{\partial f}{\partial X}(X)$$

$$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

## Depth-wise Coupling

$$\left. \begin{aligned} J_{11}^l &= \frac{\partial}{\partial x_1^{l-1}} \left( f^l(X^{l-1}) \right)_1 \\ J_{11}^{l+1} &= \frac{\partial}{\partial x_1^l} \left( f^{l+1}(X^l) \right)_1 \end{aligned} \right\} A_{11}^{l,l+1}$$

Tokens

Depth